ChatGPT

Brandes' Algorithm Explanation

explain Brandes’ Algorithm. 2
A measure of betweenness centrality based on random walks by M.E.J. Newman. 10
how brandes algorithm and A measure of betweenness centrality based on random walks by M.E.J. Newman diff. 16

explain Brandes' Algorithm

Brandes' Algorithm for Betweenness Centrality

Brandes' Algorithm is an efficient method to compute **betweenness centrality** of nodes (and optionally edges) in a graph.

It is one of the most important algorithms in network science because it reduces the computation cost dramatically compared to the naive approach.

It is commonly used in:

- Social networks
- Transportation systems
- Communication networks
- Influence analysis
- Community detection

---

1. What is Betweenness Centrality?

A node is important if it lies on many shortest paths between other nodes.

For a node v:

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

Where:

- σ_{st} = number of shortest paths from s to t
- $\sigma_{st}(v)$ = number of those shortest paths passing through v

---

Intuition

If many people/messages/data packets must pass through a node, that node controls information flow.

Example:

- Airports
- Routers
- Influencers in social networks

---

2. Problem with Naive Computation

Naively:

1. Compute shortest paths between every pair
2. Count how many pass through each node

Complexity becomes huge:

$$O(n^3)$$

or worse for sparse graphs.

For large networks this is impractical.

---

3. Brandes' Key Insight

Instead of recomputing everything independently for each pair:

- Run one shortest-path search from each source node
- Reuse information
- Accumulate dependencies efficiently

This reduces complexity to:

Unweighted Graph

$$O(VE)$$

Weighted Graph

$$O(VE + V^2 \log V)$$

Huge improvement.

---

4. Core Idea of the Algorithm

For every source node s:

1. Find shortest paths from s to all nodes
2. Count:
    - distance
    - number of shortest paths
    - predecessors
3. Process nodes in reverse shortest-path order
4. Propagate dependency values backward

5. Main Variables

For each source s:

Symbol	Meaning
$dist[v]$	Distance from source
$\sigma[v]$	Number of shortest paths from source to v
$P[v]$	Predecessors of v on shortest paths
$\delta[v]$	Dependency score

6. Forward Phase (Shortest Paths)

We perform:

- BFS for unweighted graphs
- Dijkstra for weighted graphs

During Traversal

When exploring:

First shortest path found

$$dist[w] = dist[v] + 1$$

Another shortest path found

$$\sigma[w] += \sigma[v]$$

Record predecessor

$$P[w].append(v)$$

7. Dependency Accumulation Phase

This is the genius part.

Process nodes in reverse order of distance.

Dependency formula:

$$\delta(v) = \sum_{w: v \in P[w]} \frac{\sigma(v)}{\sigma(w)} (1 + \delta(w))$$

Meaning

If:

- node w depends heavily on shortest paths through v,
- then v gets credit.

The dependency propagates backward.

---

8. Final Centrality Update

For every node except source:

$$C_B(v) += \delta(v)$$

Repeat for all sources.

---

9. Step-by-Step Example

Consider graph:

```
A --- B --- D
 \   /
   C
```

Edges:

- A-B
- A-C
- B-C
- B-D

Suppose source = A

Step 1: BFS

Distances:

Node	Distance
A	0
B	1
C	1
D	2

Step 2: Count Shortest Paths

Node	σ
A	1
B	1
C	1
D	1

Because only:

- A→B→D exists

Step 3: Reverse Processing

Process:

$$D \rightarrow C \rightarrow B \rightarrow A$$

D contributes to B

$$\delta(B) += 1$$

So B gains high centrality.

10. Why Reverse Order?

Because dependencies flow backward from farther nodes to nearer nodes.

This is similar to:

- Dynamic programming on DAGs
- Backpropagation
- Reverse topological processing

11. Complexity

Unweighted Graph

Using BFS:

$$O(VE)$$

Weighted Graph

Using Dijkstra:

$$O(VE + V^2 \log V)$$

12. Pseudocode

```
for each node s:
    initialize stack S
    initialize predecessor list P[w]
    sigma[t] = 0
    sigma[s] = 1
    dist[t] = -1
    dist[s] = 0

    BFS from s

    while queue not empty:
        pop v
        push v onto S

        for neighbor w:
            if first time visited:
                set distance

            if shortest path found:
                sigma[w] += sigma[v]
                add v to P[w]

    delta[v] = 0

    while S not empty:
        pop w

        for v in P[w]:
            delta[v] += (sigma[v]/sigma[w]) * (1 + delta[w])

        if w != s:
            BC[w] += delta[w]
```

13. Why Brandes' Algorithm is Important

Before Brandes:

- Betweenness centrality was extremely expensive.

Brandes made:

- Large-scale network analysis practical.

It became foundational in:

- Network science
- Graph mining

- Social influence analysis

14. Intuition Using Water Flow

Imagine:

- shortest paths = water pipes
- information = water flow

Nodes carrying more flow between others have:

- higher betweenness centrality.

Brandes efficiently computes how much “flow dependency” each node has.

15. Applications

Social Networks

Find influencers/brokers.

Transportation

Critical junctions.

Internet Routing

Routers with heavy traffic.

Biology

Proteins connecting modules.

Fraud Detection

Hidden intermediaries.

16. Relation to Other Centralities

Centrality	Measures
Degree	Immediate popularity

Centrality	Measures
Closeness	Distance efficiency
Eigenvector	Importance of neighbors
PageRank	Random walk importance
Betweenness	Control over shortest paths

17. Difference from PageRank

You asked earlier about random-walk betweenness vs PageRank.

They are related but different.

Feature	Brandes / Betweenness	PageRank
Based on	Shortest paths	Random walks
Focus	Path control	Probability of visit
Finds	Brokers/bridges	Popular nodes
Global flow	Deterministic shortest paths	Stochastic navigation

A node may:

- have high PageRank but low betweenness
- or vice versa.

Example:

- a bridge node between communities often has huge betweenness but modest PageRank.

18. Small Conceptual Visualization

```
Community 1 ---- X ---- Community 2
```

Node X:

- may not have many neighbors
- but all communication passes through it

Thus:

- High betweenness
- Moderate degree
- Moderate PageRank

19. Key Insight to Remember

Brandes' algorithm works because:

> shortest-path dependencies can be accumulated backward efficiently instead of recomputed repeatedly.

That is the entire breakthrough.

A measure of betweenness centrality based on random walks by M.E.J. Newman

"A Measure of Betweenness Centrality Based on Random Walks" — M.E.J. Newman

This paper by M. E. J. Newman introduced **Random Walk Betweenness Centrality**, an alternative to traditional shortest-path betweenness.

The key idea is:

> In real networks, information does not always travel along shortest paths.

Instead:

- people wander,
- packets reroute,
- influence diffuses probabilistically,
- diseases spread randomly.

So Newman proposed:

- measuring importance using **random walks** instead of only shortest paths.

---

1. Traditional Betweenness Problem

Standard betweenness assumes:

$$\text{all communication uses shortest paths}$$

But real systems rarely behave perfectly.

Examples:

- Humans do not know global shortest routes in social networks.
- Internet traffic may reroute.
- Rumors spread randomly.
- Biological signals diffuse stochastically.

So shortest-path centrality can miss important nodes.

2. Core Idea of Random Walk Betweenness

Imagine:

1. A random walker starts at source s
2. It randomly moves neighbor-to-neighbor
3. Eventually it reaches target t

A node gets high centrality if:

- many random walks between all pairs tend to pass through it.

3. Intuition

Shortest-path betweenness measures:

> "How often is this node on optimal routes?"

Random-walk betweenness measures:

> "How much natural flow diffuses through this node?"

This is often more realistic for:

- social diffusion,
- information spreading,
- biological transport.

4. Electrical Network Interpretation

This is the beautiful insight of the paper.

Newman showed:

Random walks are mathematically equivalent to electrical current flow.

Imagine:

- graph edges = resistors
- inject 1 unit current at source s
- remove current at target t

Current distributes through ALL possible paths.

Nodes carrying more current:

- have higher random-walk betweenness.

5. Difference from Shortest-Path Betweenness

Feature	Brandes Betweenness	Random-Walk Betweenness
Path assumption	Shortest paths only	All possible paths
Flow type	Deterministic	Probabilistic
Sensitive to	Geodesic structure	Diffusion structure
Realism	Efficient routing	Natural spreading
Mathematical basis	BFS/Dijkstra	Markov chains + Laplacian

6. Mathematical Foundation

Let:

- A = adjacency matrix
- D = degree matrix
- $L = D - A$ = graph Laplacian

$L = D - A$

The Laplacian governs:

- diffusion,
- random walks,
- electrical flow.

7. Random Walk Transition Probability

At node i:

Probability of moving to neighbor j:

$$P_{ij} = \frac{A_{ij}}{k_i}$$

Where:

- k_i = degree of node i

This defines a Markov chain.

8. Current Flow Interpretation

For each source-target pair (s, t):

1. Inject current at s
2. Extract at t
3. Solve voltages using Kirchhoff equations
4. Measure current through nodes

Node centrality:

$$C(v) = \sum_{s<t} I_{st}(v)$$

Where:

- $I_{st}(v)$ = amount of current flowing through node v

---

9. Why This is Powerful

Because it considers:

- ALL paths,
- weighted by probability.

Not just one shortest route.

So it captures:

- redundancy,
- diffusion,
- uncertainty,
- exploration.

---

10. Example

Consider:

```
A --- B --- D
 \         /
   \-- C--/
```

There are multiple routes from A to D:

- A-B-D
- A-C-D

Shortest-path betweenness splits equally.

But random-walk centrality also considers:

- loops,
- revisits,
- indirect wandering.

Thus the scores differ.

---

11. Relation to Electrical Current

A shortest-path method says:

> "Water only uses shortest pipes."

Random-walk/current-flow says:

> "Electricity spreads through every available conductor."

That is much closer to many real diffusion systems.

---

12. Computational Challenge

Random-walk betweenness is more expensive.

Why?

Because it requires:

- matrix inversions,
- Laplacian solving,
- linear algebra operations.

Complexity roughly:

$$O(n^3)$$

for naive implementations.

---

13. Relation to PageRank

You asked whether this is the same as PageRank.

Answer:

No — but both use random walks.

Feature	Random-Walk Betweenness	PageRank
Measures	Passage flow between pairs	Stationary visit probability
Focus	Bridge/control nodes	Popular/authoritative nodes
Source-target pairs	Yes	No
Uses all paths	Yes	Yes
Markov process	Yes	Yes
Output meaning	Mediation role	Long-term importance

14. Key Conceptual Difference

PageRank

Asks:

> "If a random surfer walks forever, where do they spend time?"

Random-Walk Betweenness

Asks:

> "Which nodes are traversed while moving between source-target pairs?"

15. Example Difference

Imagine two dense communities connected by one bridge node:

```
Cluster A ---- X ---- Cluster B
```

Node X:

- may not be visited frequently overall
- but almost every cross-community walk passes through it

Thus:

- HIGH random-walk betweenness
- maybe LOW PageRank

16. Connection to Diffusion Models

This idea strongly connects to:

- epidemic spreading,
- diffusion processes,
- influence maximization,
- information propagation.

Especially in:

- SIS/SIR models,
- network flow,
- community bridging.

---

17. Why This Paper Was Important

The paper helped shift network science from:

- purely shortest-path thinking

toward:

- diffusion-based thinking.

This influenced:

- spectral graph theory,
- current-flow centrality,
- resistance distance,
- graph neural diffusion methods.

---

18. Main Takeaway

Traditional betweenness says:

> "Important nodes lie on shortest paths."

Newman's random-walk betweenness says:

> "Important nodes are those through which diffusion naturally flows."

That is a much richer and often more realistic notion of influence in complex networks.

how brandes algorithm and A measure of betweenness centrality based on random walks by M.E.J.

Difference Between Brandes' Algorithm and Newman's Random-Walk Betweenness

These two are closely related because both deal with **betweenness centrality**, but they are fundamentally different in:

- what they measure,
- the assumptions they make,
- the mathematics behind them,
- and how they are computed.

---

1. The Core Difference

Aspect	Brandes' Algorithm	Newman's Random-Walk Betweenness
Measures	Shortest-path betweenness	Random-walk/current-flow betweenness
Assumption	Information uses shortest paths	Information diffuses randomly
Path usage	Only geodesics	All possible paths
Flow type	Deterministic	Probabilistic
Main mathematics	BFS/Dijkstra + DP	Markov chains + Laplacian
Goal	Efficient computation	More realistic diffusion model

---

2. Important Clarification

Brandes is an ALGORITHM

It is a fast method to compute:

$$\text{Shortest-path betweenness centrality}$$

Newman's method is a DIFFERENT CENTRALITY DEFINITION

It defines:

$$\text{Random-walk betweenness centrality}$$

So:

- Brandes = efficient computation technique

- Newman = alternative notion of importance

They are not competing algorithms for the same thing.

---

3. What Brandes Assumes

Brandes assumes:

> Communication always travels along shortest paths.

So if there are two shortest paths:

- flow splits among them,
- longer paths are ignored.

Example

```
A --- B --- D
 \         /
   \-- C--/
```

From A to D:

- A-B-D
- A-C-D

Brandes considers ONLY these shortest paths.

It ignores:

- A-B-A-C-D
- A-C-A-B-D
- loops
- wandering paths

---

4. What Newman Assumes

Newman assumes:

> Movement behaves like a random walk.

A walker:

- randomly chooses neighbors,
- may wander,
- may revisit nodes,

- eventually reaches target.

Thus ALL paths contribute.

Even long weird paths matter (with lower probability).

---

5. Philosophical Difference

Brandes / Shortest-Path Betweenness

Models:

- optimal routing,
- rational navigation,
- efficient communication.

Good for:

- road routing,
- engineered systems,
- optimal transport.

Newman / Random-Walk Betweenness

Models:

- diffusion,
- uncertainty,
- exploration,
- stochastic spreading.

Good for:

- rumors,
- epidemics,
- biological processes,
- social diffusion.

---

6. Mathematical Difference

Brandes

Uses shortest-path counting:

$$C_B(v) = \sum_{s,t} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

Where:

- σ_{st} = shortest paths
- $\sigma_{st}(v)$ = shortest paths through v

Newman

Uses current flow / random walks.

Based on:

- graph Laplacian,
- electrical networks,
- Markov processes.

Core matrix:

$L = D - A$

where:

- L = graph Laplacian.

---

7. Computation Difference

Brandes

Uses:

- BFS (unweighted)
- Dijkstra (weighted)

Then:

- dynamic programming backward accumulation.

Complexity:

$$O(VE)$$

for unweighted graphs.

Efficient even for large networks.

Newman

Requires:

- solving linear systems,
- matrix inversion,
- current-flow equations.

Complexity often near:

$$O(n^3)$$

Much heavier computationally.

---

8. Example Where They Differ

Consider:

```
Cluster A ---- X ---- Cluster B
```

Node X is the only bridge.

Both methods give X high importance.

Now add many alternate longer paths.

```
A == many indirect routes == B
```

Brandes

Still focuses mostly on shortest paths.

Newman

Recognizes:

- diffusion through indirect routes,
- probabilistic wandering,
- flow spreading everywhere.

So scores change substantially.

9. Flow Interpretation

Brandes

Like cars using:

- GPS shortest routes.

Newman

Like electricity or heat:

- spreading through every available path.

10. Sensitivity to Network Structure

Property	Brandes	Newman
Sensitive to geodesics	Very high	Moderate
Sensitive to redundancy	Less	Strong
Captures diffusion	Weakly	Strongly
Captures uncertainty	No	Yes

11. Relation to PageRank

Newman's method is closer to PageRank than Brandes is because both use:

- random walks,
- stochastic movement,
- diffusion ideas.

But:

	Newman	PageRank
Measures	Passage between pairs	Long-run visit probability
Source-target specific	Yes	No
Finds	Bridges	Popular nodes

12. Real-World Analogy

Brandes

Imagine commuters:

- always choosing fastest routes.

Newman

Imagine rumors:

- spreading unpredictably through conversations.

---

13. Which One is Better?

Depends on the system.

Use Brandes when:

- shortest paths matter,
- routing is optimized,
- graph is huge,
- efficiency matters.

Examples:

- transportation,
- routing,
- communication networks.

Use Newman when:

- diffusion matters,
- movement is uncertain,
- spreading is stochastic.

Examples:

- epidemics,
- information diffusion,
- biological signaling,
- social influence.

---

14. Deep Insight

These methods represent two different views of networks:

Brandes View

Networks are:

- routing systems.

Importance = path control.

Newman View

Networks are:

- diffusion media.

Importance = flow mediation.

---

15. Final One-Line Summary

Brandes

> "How often is a node on shortest paths?"

Newman Random-Walk Betweenness

> "How much probabilistic flow naturally passes through a node?"